

MANAGEMENT REVIEW

Management without theatre.

KPIS

DASHBOARDS THAT GET USED

Few's one screen at a glance, how sure people feel with data, the bullet graph, who uses dashboards and for what, five design mistakes, and a card for a dashboard that gets used.



21%

of 9,000 employees in nine countries said they were confident in their data literacy skills.

ACCENTURE & QLIK, 2020

INSIDE

COVER STORY

One screen, one glance

PAGE 03

THE MODEL

A line instead of a gauge

PAGE 05

TOOL OF THE ISSUE

The dashboard card

PAGE 08

NO. 37 · KPIS

In this issue

Many companies have more dashboards than decisions. This issue asks what a dashboard is for, who can read it, who actually uses it, and how to tell a screen that gets used from one that is merely switched on.

Stephen Few's definition is narrow: the most important information on one screen, monitored at a glance. In a 2019 survey of 9,000 employees in nine countries, 21% were confident in their data literacy and 74% felt overwhelmed or unhappy working with data. In Finland only about a quarter of the sales managers who answered a 2010 survey used a dashboard. Few's bullet graph replaces the gauge with one line, and a review of health care studies links dashboards that are easy to reach with better care.

Stiven Janaqi EDITOR · STIVEN CATALYST

03 COVER STORY
One screen, one glance

04 THE NUMBERS
Data that overwhelm

05 THE MODEL
A line instead of a gauge

06 WHAT THE RESEARCH SAYS
Who uses them, and for what

07 HOW IT IS MEASURED
Is anyone looking?

08 TOOL OF THE ISSUE
The dashboard card

09 SOURCES
Sources and method

HOW TO READ THIS ISSUE

Figure

Every figure has its source and year at the foot of its page.

Our reading

Where the editors interpret rather than the research, it says so.

Practice

The steps and the card are proposals to try, not research results.

COVER STORY

One screen, one glance

In 2004 Stephen Few defined a dashboard narrowly: a visual display of the most important information needed to reach one or more objectives, arranged on a single screen so it can be monitored at a glance.

01**The most important**

what the objectives need, not everything that can be measured

02**One screen**

no scrolling, no switching: short-term memory holds little

03**At a glance**

summaries and exceptions, read in an instant

Later Few sharpened it: a mainly visual display that people use to monitor current conditions rapidly, conditions that need a timely response in a specific role. By that test, he wrote in 2017, only 2 of the 28 examples in *The Big Book of Dashboards* were rapid-monitoring displays; the others may be useful, but they serve other purposes.

OUR READING

A screen full of charts is not yet a dashboard. Ask who looks at it, how often, and what they do next.

Sources: Stephen Few, *Perceptual Edge*, 2017; Stephen Few, *O'Reilly Media* (2nd ed. 2013, Analytics Press), 2006 (via UXmatters, 2007)

The definitions are Few's; the 2004 wording was checked where he quotes it himself in 2017. The count of 2 in 28 is his judgement of another book, not a measurement.

THE NUMBERS

Data that overwhelm

In September 2019 Opinium surveyed 9,000 full-time employees in nine countries, Germany among them, for Accenture and Qlik. Most saw data as an asset; few felt sure using it.

EMPLOYEES IN NINE COUNTRIES, SEPTEMBER 2019 (SELF-REPORTED)

See data as an asset



Overwhelmed or unhappy working with data



Often trust a gut feeling instead of data



Confident in their data literacy



In a 2022 Forrester Consulting survey for Tableau of more than 2,000 people in ten countries, 82% of decision-makers expected basic data literacy in every department, but only 39% of organisations offered data training to all employees.

OUR READING

Dashboards are read by people, not by software. A clear screen does not help someone who avoids numbers.

Sources: Accenture & Qlik for The Data Literacy Project, survey by Opinium, 2020; Forrester Consulting, commissioned by Tableau, 2022 (via Help Net Security, 2022)

Both studies were commissioned by firms that sell analytics software, and both rest on self-reports. For Albania we have no comparable figure.

THE MODEL

A line instead of a gauge

Few designed the bullet graph to replace the meters and gauges common on dashboards. In a small space it shows one measure, compares it with a target and places it in ranges such as poor, satisfactory and good.

01 Label names the measure	02 Scale one linear axis, usually from zero	03 Measure a dark bar, the most visible part	04 Comparison a short cross line: target or last year	05 Ranges two to five shades, ideally three
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Few advises shades of one colour, dark for poor and light for good, rather than separate hues that people with colour blindness may not tell apart. A linear design, he writes, also takes less space and is read more efficiently than a round gauge.

HYPOTHETICAL EXAMPLE, ONE LINE ON A SHIFT DASHBOARD

Measure	on-time departures this week: 91%
Target	a cross line at 95%
Ranges	below 85% poor, 85–95% satisfactory, above 95% good

One line: below target, in the middle range. The figures are invented.

Source: Stephen Few, Perceptual Edge, 2013

The parts and defaults follow Few's design specification (2006, revised 2013). The colour advice is his recommendation, not a tested standard.

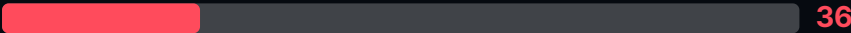
WHAT THE RESEARCH SAYS

Who uses them, and for what

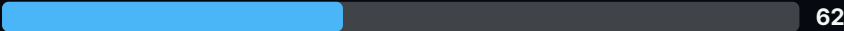
At the end of 2010 Oana Velcu-Laitinen and Ogan Yigitbasioglu surveyed senior sales managers at 851 Finnish companies about dashboards. 145 answered, and only about a quarter of them used one.

SALES MANAGERS WHO ANSWERED, FINLAND, END OF 2010 (NUMBER, OF 145)

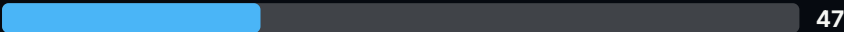
Used a dashboard



Used other reporting tools instead



Other non-users



Seven of the 36 had built theirs in Excel. Users ranked communication and consistency first of four purposes, ahead of monitoring, and data quality went with use. A 2015 review of 11 health care studies found that where clinicians could reach dashboards easily, as a screen saver for example, use went with better care processes and patient outcomes.

OUR READING

A dashboard people have to go looking for is rarely looked at. Put it where the work happens.

Sources: Oana Velcu-Laitinen & Ogan M. Yigitbasioglu, International Journal of Digital Accounting Research, 2012; Dawn Dowding et al., International Journal of Medical Informatics, 2015

A small sample (17% responded) and correlations, not proof of cause; 47 is the rest of the 109 non-users. The health care review calls the evidence limited.

HOW IT IS MEASURED

Is anyone looking?

A dashboard can be checked twice: against Few's list of common design mistakes, and against what people do with it. The first takes an hour; the second takes a month of meetings.

Five of Few's thirteen common mistakes

1. exceeding the boundaries of one screen
2. too little context: no target, no comparison
3. excessive detail or precision
4. a deficient measure
5. misusing or overusing colour

1

Name the user and the decision

If no one can say which decision a number serves, it belongs in a report.

2

Count actions, not views

In each review, note which number led to a step, an owner and a date.

3

Remove what no one used

A measure that triggered nothing for a quarter leaves the screen.

Source: Stephen Few, O'Reilly Media (2nd ed. 2013, Analytics Press), 2006 (via UXmatters, 2007)

The mistakes are from Few (2006); the steps are the editors' proposal, not a tested method.

[More in the essay](#)

KPIs do not improve in Excel

TOOL OF THE ISSUE

The dashboard card

Fill it in before you build the screen, and again after a month. If a line stays empty, the dashboard is not ready.

01 Who looks at it the role, and how often: every shift, day or week

02 The decision it serves one sentence: what changes when the number moves

03 Three to five measures each with its definition and its source

04 The comparison target, last week or last year, for every measure

05 When it is read the meeting or the moment, and who leads it

06 What triggers action the threshold, who acts, by when

07 What we removed measures that led to no action in a month

Sources: Stephen Few, *Perceptual Edge*, 2017; Stephen Few, *O'Reilly Media* (2nd ed. 2013, *Analytics Press*), 2006 (via *UXmatters*, 2007)

A practice proposed by the editors, after Few's definition of a dashboard (2004, 2017) and his book (2006).

Open the tool: [Shift Pulse](#)

stivencatalyst.com/tools/shift-pulse/

SOURCES AND METHOD

Every figure has a source.

The figures in this issue come from the sources below. The year shows how recent each one is.

Information Dashboard Design: The Effective Visual Communication of Data

Stephen Few, O'Reilly Media (2nd ed. 2013, Analytics Press), 2006 · via UXmatters, 2007

There's Nothing Mere About Semantics

Stephen Few, Perceptual Edge, 2017
<https://www.perceptualedge.com/blog/?p=2793>

The Human Impact of Data Literacy

Accenture & Qlik for The Data Literacy Project, survey by Opinium, 2020

<https://www.qlik.com/us/news/company/press-room/press-releases/data-skills-gap-is-costing-organizations-billions-in-lost-productivity>

Building Data Literacy: The Key to Better Decisions, Greater Productivity, and Data-Driven Organizations

Forrester Consulting, commissioned by Tableau, 2022 · via Help Net Security, 2022

Bullet Graph Design Specification

Stephen Few, Perceptual Edge, 2013

https://www.perceptualedge.com/articles/misc/Bullet_Graph_Design_Spec.pdf

The Use of Dashboards in Performance Management: Evidence from Sales Managers

Oana Velcu-Laitinen & Ogan M. Yigitbasioglu, International Journal of Digital Accounting Research, 2012

https://doi.org/10.4192/1577-8517-v12_2

Dashboards for improving patient care: Review of the literature

Dawn Dowding et al., International Journal of Medical Informatics, 2015

<https://eprints.whiterose.ac.uk/84905/>

EDITORIAL METHOD

Each figure was checked for its year, its publisher and what exactly it measures. Where the publisher's page could not be opened, the figure was checked against independent summaries and is marked “via”. The editors' interpretation is marked “Our reading”. Figures that could not be confirmed are not in the issue.



MANAGEMENT REVIEW

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Every issue, one management question, checked against the best research.

ALL ISSUES

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